

Probability Theory Notes

Urn / Matching Problems

Derangements

$$D_m = m! \sum_{j=0}^m \frac{(-1)^j}{j!} \quad \text{Q1}$$

$$P(X = k) = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}$$

As $n \rightarrow \infty$:

$$\sum_{j=0}^{\infty} \frac{(-1)^j}{j!} = e^{-1} = \frac{1}{e}$$

$$\boxed{\lim_{n \rightarrow \infty} P(X = k) = \frac{1}{e k!}}$$

Polya Urn — Draw 2 (with replacement + c extra)

Setup: urn has a red and b blue balls, $N = a + b$.

Q2

$$P(R_1) = \frac{a}{a+b}, \quad P(B_1) = \frac{b}{a+b}$$

$$P(R_2) = P(R_2 | R_1)P(R_1) + P(R_2 | B_1)P(B_1)$$

$$P(R_2 | R_1) = \frac{a}{a+b+c}, \quad P(R_2 | B_1) = \frac{a+c}{a+b+c}$$

$$\therefore P(R_2) = \frac{a(a)}{(a+b+c)(a+b)} + \frac{(a+c)(b)}{(a+b+c)(a+b)} = \frac{a(a+b) + bc}{(a+b)(a+b+c)}$$

Polya Urn — Draw 3

$N = a + b$. Possible second-draw outcomes: RR, BR, RB, BB.

Q3

$$P(RR) = \frac{a}{N} \cdot \frac{a}{N+c}, \quad P(R_3 | RR) = \frac{a}{N+2c}$$

$$P(BR) = \frac{b}{N} \cdot \frac{a+c}{N+c}, \quad P(R_3 | BR) = \frac{a+c}{N+2c}$$

$$P(RB) = \frac{a}{N} \cdot \frac{b+c}{N+c}, \quad P(R_3 | RB) = \frac{a(b+c)(a+c)}{N+2c} \quad (\text{to be multiplied})$$

$$P(BB) = \frac{b}{N} \cdot \frac{b}{N+c}, \quad P(R_3 | BB) = \frac{a+2c}{N+2c}$$

$$\therefore P(R_3) = \frac{a^3 + b(a+c)^2 + a(b+c)(a+c) + b^2(a+c)}{N(N+c)(N+2c)}$$

Section B

Prisoner Problem

$P(A) = P(B) = P(C) = \frac{1}{3}$. Let W be the event that B is executed.

Q7

$$P(W | A) \rightarrow A \text{ is pardoned} \Rightarrow B \text{ or } C \rightarrow \text{execute} \Rightarrow \frac{1}{2}$$

$$P(W | B) \rightarrow B \text{ is pardoned} \Rightarrow A \text{ or } C \rightarrow \text{execute} \Rightarrow 0$$

$$P(W | C) \rightarrow C \text{ is pardoned} \Rightarrow A \text{ or } B \rightarrow \text{execute} \Rightarrow 1$$

$$P(W) = P(W | A)P(A) + P(W | B)P(B) + P(W | C)P(C) = \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$$P(A | W) = \frac{P(W | A)P(A)}{P(W)} = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3} \quad P(C | W) = \frac{2}{3}$$

Geometric-like — First Success

$$P(A) = p + (1-p)^2p + \dots$$

Q8

$$= p[1 + (1-p)^2 + \dots] = \frac{p}{1 - (1-p)^2} = \frac{p}{2p - p^2} = \frac{1}{2-p}$$

$\therefore g(A) = \frac{1}{2-p}$. If $p = 0 \Rightarrow$ undefined; but $p \in (0, 1) \Rightarrow$ no problem.

Section C — Independence

Note. A, B, C are independent $\Rightarrow P(A \cap B) = P(A)P(B)$ and $P(A \cap C) = P(A)P(C)$. **Q9**
Counterexample:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}, \quad A = \{1, 2, 3, 4\}, \quad B = \{1, 2, 3, 4\}, \quad C = \{1, 5, 6, 7\}$$

$$\Omega = \{1, 2, 3, 4\}, \quad P(A \cap B \cap C) = \frac{1}{8} \quad \text{but} \quad P(A) = P(B) = P(C) = \frac{1}{2}$$

So A, B, C are *not* mutually independent (though pairwise independent).

Section D — CDF and Jumps

Jumps of CDF / Countability of Discontinuities

$F_X(x) = P(X \leq x)$; jump at x :

Q14

$$\Delta(x) = F_n(x) - F_n(x^-) = P(X \leq x) - P(X < x) = P(X = x)$$

If F_n is discontinuous $\Rightarrow P(X = x) > 0$.

For each $n \geq 1$ let $D_n = \{x : \Delta(x) \geq \frac{1}{n}\}$.

$$\therefore P(X = x_i) \geq \frac{1}{n} \quad (i = 1, \dots, m)$$

Since events are disjoint:

$$1 \geq P\left(\bigcup_{i=1}^m \{X = x_i\}\right) = \sum_{i=1}^m P(X = x_i) \geq \frac{m}{n} \quad \Rightarrow \quad m \leq n \quad \Rightarrow \quad |D_n| \leq n \text{ (finite)}$$

$$\therefore \Delta x = P(X = x) > 0 \Rightarrow 1/n < \Delta(x) \text{ for some } n \Rightarrow x \in D_n$$

$$\therefore D = \bigcup_{n=1}^{\infty} D_n \text{ (countable)}$$

Discrete Distribution Dense in $(0, 1)$

$$F(x) = \sum_n \frac{1}{2^n}, \quad \sum_{n=1}^{\infty} \frac{1}{2^n} = 1, \quad P(X = q_n) = 2^{-n} > 0 \text{ (discrete)}$$

Q111

For $b > a$ in $(0, 1)$: $a < q_n \leq b \Rightarrow$ dense, action needed.

$$F(b) - F(a) = P(X = q_n) = \frac{1}{2^n} > 0$$

Section E

PDF Proof — Unbounded Example

A PDF must satisfy: $f(x) \geq 0 \forall x$ and $\int_{-\infty}^{\infty} f(x) dx = 1$.

Q1

False/tricky example:

$$f(x) = \frac{1}{2\sqrt{x}}, \quad 0 \leq x \leq 1; \quad 0 \text{ otherwise.}$$

$$\int_0^1 \frac{1}{2\sqrt{x}} dx = 1 \quad \checkmark$$

But as $x \rightarrow 0^+$, $f(x) \rightarrow \infty \Rightarrow f$ is unbounded (yet still a valid PDF).

Independent RVs with Same Continuous CDF

X_1, X_2 independent with the same continuous CDF $F(y)$.

Q5

$$P(X_1 = X_2 \mid X_2 = y) = P(X_1 = y) \xrightarrow{\text{independence}} 0$$

$$P(X_1 = X_2) = \int P(X_1 = X_2 \mid X_2 = y) dF(y) = 0 \quad (\text{only } 0 \text{ when } P(X_1 = y) = 0)$$

(Doubt)

Mixed Distribution — Quantile

Q6

$P(X = 0) = p$ where $0 < p < 1$; with probability $1-p$, X follows a continuous distribution with strictly increasing CDF $G(x)$ on $x > 0$. Q7

$$F(x) = \begin{cases} 0 & x < 0 \\ p & x = 0 \\ p + (1-p)G(x) & x > 0 \end{cases}$$

q -th quantile: $\xi_q = \inf\{x : F(x) \geq q\}$.

Case 1: $0 \leq q \leq p$

$F(0) = p \geq q \Rightarrow$ smallest x with $F(x) \geq q$ is $x = 0$.

$$\therefore \xi_q = 0 \quad \text{for } 0 \leq q \leq p$$

Case 2: $p < q < 1$

$F(x) = q$, i.e. $p + (1-p)G(x) = q$.

$$G(x) = \frac{q-p}{1-p}$$

(strictly increasing in x)

$$\therefore \xi_q = G^{-1}\left(\frac{q-p}{1-p}\right), \quad p < q < 1$$

Summary:

$$\xi_q = \begin{cases} 0 & 0 \leq q \leq p \\ G^{-1}\left(\frac{q-p}{1-p}\right) & p < q < 1 \end{cases}$$

Transformation of Random Variables

Uniform on a Disk of Radius R

$$F_R(\delta) = P(X \leq \delta) = \frac{\text{area of disk of radius } \delta}{\text{total area}} = \frac{\pi\delta^2}{\pi R^2} = \frac{\delta^2}{R^2}$$

Q12

$$F_R(\delta) = \begin{cases} 0 & \delta < 0 \\ \delta^2/R^2 & 0 \leq \delta \leq R \\ 1 & \delta > R \end{cases}$$

$$f_R(\delta) = \frac{d}{d\delta} F_R(\delta) = \begin{cases} 2\delta/R^2 & 0 \leq \delta \leq R \\ 0 & \text{otherwise} \end{cases}$$

Verification:

$$\int_0^R \frac{2\delta}{R^2} d\delta = \frac{2}{R^2} \cdot \frac{R^2}{2} = 1 \quad \checkmark$$

$$X \sim \text{Uniform}(-1, 1), Y = X^2$$

Since $-1 < X < 1 \Rightarrow 0 \leq Y < 1$.

Q13

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$$

Since X is uniform on $(-1, 1)$:

$$P(a \leq X \leq b) = \frac{b-a}{2}$$

$$F_Y(y) = \frac{\sqrt{y} - (-\sqrt{y})}{2} = \sqrt{y}$$

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ \sqrt{y} & 0 \leq y < 1 \\ 1 & y \geq 1 \end{cases}$$

$$f_Y(y) = \frac{d}{dy} \sqrt{y} = \frac{1}{2\sqrt{y}} \quad \text{Verification: } \int_0^1 \frac{1}{2\sqrt{y}} dy = 1 \quad \checkmark$$